T|W|I
Thurgauer
Wirtschaftsinstitut
an der Universität Konstanz

When personalization deters delegation to algorithms: Experimental evidence

Lara Suraci & Irenaeus Wolff

— 23rd June, 2026, Symposium on 'AI and the Future of Work', Berlin —

Background

- Huge & unconnected literature on algorithm/automation/AI/... reliance (→ review by Chugunova & Sele, 2022)
- Most of the literature: setups with clearly identifiable mistakes
- But in many real-life settings, preferences determine what is a mistake
- E.g., robo-advisors: maximal payoff *vs* best fit to risk-parameters

Relevance?

Well-known (Swiss) robo-advisors(' marketing strategies)

- "We rely on a low-cost, passive investment strategy and have selected a total of 9 equity, bond and real estate ETFs from over 1,500 ETFs (exchange-traded funds) and put them together in such a way that they achieve an optimal risk/return ratio." (findependent.ch; for GER, see e.g., whitebox.eu)
- "Chat with Selma: Selma looks at your financial life and learns how you should structure your finances. Get your plan: Selma creates a personalized investment plan that fits your life. Sit back and relax! Selma takes care of your money from now on and does all the buying and selling for you." (selma.com; for GER, see e.g., quirion.de)

Research Question

- Does algorithm reliance differ for **objective optimization *vs.* preference-based personalisation** in a preferential-choice environment?
    (i) ...at first contact?
    (ii) ...overall?

Objective vs Personalised Algorithms

- What do we mean by **objective optimization**?
  - an algorithm that chooses some 'objectively ideal' option independently of a subject's preferences (here: **highest EV**)
- What do we mean by **personalised**?
  - an algorithm that measures a subject's preferences and chooses accordingly (here: **implementation of revealed preferences**)

Objective vs Personalised Algorithms

T|W|I
Thurgauer
Wirtschaftsinstitut
an der Universität Konstanz

- What do we mean by **objective optimization**?
  - an algorithm that chooses some 'objectively ideal' option independently of a subject's preferences (here: **highest EV**)
- What do we mean by **personalised**?
  - an algorithm that measures a subject's preferences and chooses accordingly (here: **implementation of revealed preferences**)
- In the experiment, we describe both algorithms in intuitive terms:

EV: *'The algorithm works out how much each option would yield on average and selects the option for which this is highest.'*

Prefs: *'The algorithm analyses your choices from Part 1 and selects the best option for you in light of your Part-1 choices.'*

Experimental Design

Part 1 – Elicitation of preferences

- ordered lotteries, no foils, unlimited time
- “hypothetical, but the choices may be meaningful for Part 2”

Part 2 – Lottery Selection Task

- could stand for an investment task
- complexity (number of options, outcomes under different SOTW)
- time pressure (think: relative to the number of options)

Tasks

	State 1	State 2	State 3	State 4	State 5	State 6	State 7	State 8	State 9	State 10
Option 1	225	225	225	225	225	225	225	205	0	0
Option 2	525	525	525	525	525	0	0	0	0	0

State:	1	2	3	4	5	6	7	8	9	10
Option 1	225	225	0	225	0	225	225	225	225	0
Option 2	525	0	505	0	525	0	0	0	525	505
Option 3	525	0	525	0	525	0	0	0	525	525
Option 4	225	225	0	225	205	205	205	225	225	0
Option 5	225	205	0	205	205	225	225	205	225	0
Option 6	225	225	0	225	205	225	225	225	225	0
Option 7	525	0	525	0	525	0	0	0	225	525
Option 8	505	0	525	0	505	0	0	0	505	525

- For further distraction: 4 filler tasks in between
- 8 rounds: 5’ pre-view — 5’ algorithm choice—8’ decision/waiting time

Tasks

Thurgauer Wirtschaftsinstitut
an der Universität Konstanz

	State 1	State 2	State 3	State 4	State 5	State 6	State 7	State 8	State 9	State 10
Option 1	225	225	225	225	225	225	225	205	0	0
Option 2	525	525	525	525	525	0	0	0	0	0

State:	1	2	3	4	5	6	7	8	9	10
Option 1	225	225	0	225	0	225	225	225	225	0
Option 2	525	0	505	0	525	0	0	0	525	505
Option 3	525	0	525	0	525	0	0	0	525	525
Option 4	225	225	0	225	205	205	205	225	225	0
Option 5	225	205	0	205	205	225	225	205	225	0
Option 6	225	225	0	225	205	225	225	225	225	0
Option 7	525	0	525	0	525	0	0	0	225	525
Option 8	505	0	525	0	505	0	0	0	505	525

- For further distraction: 4 filler tasks in between
- 8 rounds: 5’ pre-view — 5’ algorithm choice—8’ decision/waiting time

Pre-registered analyses

Thurgauer
Wirtschaftsinstitut
an der Universität Konstanz

- First-contact delegation (11%ge pts, $p = 0.041$)
- Delegation over all rounds (no.)